

# THE EFFECTS OF FERAL GOATS (*CAPRA HIRCUS*) ON ISLAND ECOSYSTEMS

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## ABSTRACT

*Feral goats (Capra hircus) are implicated in habitat destruction, and alteration of species composition on sensitive insular ecosystems. In the absence of population control, goats have become the ecologically dominant species on many islands with the result that numerous endemic plant species have been extirpated, or are threatened by excessive grazing. It is demonstrated that removal of goats can lead to rapid recovery of suppressed vegetation. The problems associated with excessive numbers of feral goats have rarely been studied or formally recognised. Extensive and intensive research is critically needed if affected island ecosystems are to be preserved or restored.*

## INTRODUCTION

One of the more serious ecological problems in many areas, especially on oceanic islands, is habitat degradation caused by feral herbivorous animals. I define feral animals as those that: (1) exist in a wild and free-roaming state, (2) whose populations are self-sustaining, and (3) whose ancestors went through a prolonged period of domestic existence before becoming wild.

Virtually any domestic species has the potential for a feral existence; however, some such as goats (*Capra hircus*) and pigs (*Sus scrofa*) are particularly well suited. These animals are both extreme generalists and are capable of existing in a wide variety of climates and habitat types, usually having some serious effect on the native biota. Feral sheep (*Ovis aries*) occur in about a dozen locations in New Zealand (Whitaker, 1976), and on St Kilda, an archipelago to the west of Scotland, where they have been intensively studied (Jewell *et al.*, 1974). In spite of some recorded

changes in the vegetation on islands having feral sheep, there is no clear evidence that the sheep alone have been responsible for these changes. The other feral herbivores of importance are the horse (*Equus caballus*) and the burro (*Equus asinus*), but these species are of serious consequence only in western North America. Feral carnivores such as the dog (*Canis familiaris*) and the domestic cat (*Felis domesticus*) also have an extensive worldwide distribution, and can have severe detrimental effects upon native wildlife species, as for example, predation on the eggs and young of the Galapagos tortoise (*Geochelone elephantopus*) by dogs (Hendrickson, 1966; MacFarland, 1974a,b).

The objectives of this paper are: (1) to summarise what is known about the effects that goats have had upon insular ecosystems, and (2) to point out the need for further research so that they might be effectively and efficiently controlled.

#### HISTORY OF GOAT INTRODUCTIONS

Populations of feral goats have resulted either from animals that escaped accidentally or from deliberate liberations. Establishment of goats on many oceanic islands stems from introductions by whalers and other seafarers of the 17th and 18th centuries. The purpose of these introductions was to provide a ready supply of fresh meat for future visits, a worthwhile goal considering the poor nutrition available aboard ships in those days.

Very few cases of accurate documentation of either the date of liberation or origin of goats exist. For example, on Santa Catalina Island, California, it is unclear whether goats were liberated by the Spanish in the 17th century or the English in the 18th century. The first known record of goats on Santa Catalina Island was made in 1827 (Dunkle, 1950), and they were established at that time. On Ascension Island, the French introduced goats sometime prior to 1701, when they were reported to be common (Duffey, 1964).

The history of feral goats in the Galapagos Archipelago is likewise shrouded in uncertainty. The date of original introduction could be anytime from before 1700 until nearly 1835, when Darwin observed goats on Charles Island (Isla Santa Maria) (Koford, 1966). Unlike on most oceanic islands, the introductions of goats into the Galapagos were not all in the early days of exploration. Goats were initially introduced on to Isla Pinta (Abingdon Island) in 1959, where they rapidly increased to at least 20,000 individuals by 1971 (Hamann, 1975).

The history of the first introductions of goats to both New Zealand and Hawaii is known. Captain Cook released goats along the coast of New Zealand in 1773 and landed goats on at least two Hawaiian Islands in 1778 (Baker & Reeser, 1972). Goats are now firmly established on both island systems, and there appears to be little possibility of completely exterminating them, at least from New Zealand (Rudge, 1976). The island of St Helena suffered what was probably the earliest recorded introduction of goats. Wallace (1911) reported that goats were introduced to the

island in 1513 by the Portuguese, and that by 1588 they were numbered in thousands.

#### EFFECTS UPON INSULAR BIOTA

The biota of oceanic islands has in most cases evolved over long periods of isolation from mainland faunas. Since large herbivores were presumably unable to cross large distances over water, the plants that evolved on oceanic islands either lost adaptations which might have protected them from herbivores, or failed to develop new ones. Introductions of goats on to these oceanic islands resulted in levels of grazing pressure far in excess of what the plants were adapted to withstand. Either the endemic plants succumbed outright, or they were unable to compete with more vigorous exotic plants that could better withstand the high grazing and browsing intensity. As a result, many plant species disappeared, and many more were severely limited in range and number.

This assertion has been verified in several instances. The island of St Helena is a classic example of the destructive ability of goats (Wallace, 1911; Wodzicki, 1950). Originally it was heavily forested, but as the trees were felled by man, seedling regeneration was prevented by the browsing activities of the goats. By 1810, St Helena had become a barren, rocky wasteland (Wodzicki, 1950).

Yocom (1967) stated that goats in Haleakala National Park, Hawaii, eliminated some species of plants and greatly altered the abundance and distribution of others. Baker & Reeser (1972) presented a list of 19 plants considered to be rare and endangered in Haleakala, and partly attributed their rarity to over-browsing by feral goats. One of these plant species (*Canavalia* sp.) existed only in one goat enclosure and was an undescribed species when it first appeared after construction of the enclosure in 1969. Baker & Reeser (1972) surmised that the seeds of this plant may have lain dormant for as long as 100 years, germinating only after the goats were removed, and mulch build-up caused an increase in soil moisture.

In the Galapagos, feral mammals, particularly the goat, have been identified as the greatest threat to the endemic vegetation (Simkin *et al.*, 1972). Endemic tree cacti (*Opuntia* sp.) were also destroyed by goats (Dawson, 1966), and great differences have been described between islands with and without goats (Koford, 1966). The changes in vegetation communities have been described for New Zealand (Atkinson, 1964; Sykes, 1969), and at least 48 indigenous and 18 introduced plant species have disappeared from Santa Catalina Island in recent times, probably partially due to the large populations of goats (Thorne, 1967).

Goats have had severe effects on the total density of vegetation, in addition to their direct effects on the composition of plant communities. The removal of vegetation by them has been so severe in many areas that extensive erosion has occurred along with an associated decrease in soil fertility and moisture retention (Coblentz, 1974).

Large-scale habitat degradation by goats has affected endemic animal species as well as the plants. The feral goat was a competitor of the nene (*Branta sandvicensis*) in Hawaii, and a threat to the re-establishment of these endangered geese (Yocom, 1967). Baker & Reeser (1972) attributed the decline of many honeycreepers (Drepaniidae) to forest destruction, which to a large extent was caused by goats. Goats have been identified as direct competitors for food with Galapagos tortoises (Hendrickson, 1966; MacFarland, 1974a,b).

On Santa Catalina Island, goats have so severely altered the vegetation of some areas that several species of snake (*Thamnophis* sp., *Crotalus* sp., *Lampropeltus* sp.) were virtually absent. The endemic race of valley quail (*Lophortyx californicus*) was noticeably less abundant in areas of high goat density, as were the channel island fox (*Urocyon litoralis*), and the two native mice (*Peromyscus* sp. and *Peithrodontomys* sp.) (Coblentz, 1976). Both the introduced mule deer (*Odocoileus hemionus*) and bison (*Bison bison*) rarely entered areas of high goat density, presumably because of the lack of high quality forage (Coblentz, 1974). The effect that habitat destruction by goats has had upon passerine birds, on Santa Catalina and elsewhere, is largely unknown. However, Greenway (1958) has attributed the extinction of three endemic subspecies of small birds on Guadalupe Island, Mexico, to habitat destruction by 'an enormous herd of hungry goats.' The avifaunas on other goat-inhabited islands are probably also adversely affected.

Virtually all that has been written on the ecological effects of feral goats has been descriptive; only a few studies have been quantitative. Mueller-Dombois & Spatz (1972, cited in Baker & Reeser, 1972) found that percent cover of vegetation inside exclosures in goat habitat in Hawaii increased in comparison with non-exclosures areas. There was also a notable regeneration of woody species on Isla Santa Fe (Barrington) in the Galapagos, attributable to the complete removal of goats from the island (Hamann, 1975). Feral goats in Hawaii Volcanoes National Park prevented replacement of koa trees (*Acacia koa*) by browsing virtually all available suckers between 0.5 and 2 m in height (Spatz & Mueller-Dombois, 1973).

On Santa Catalina Island, I found that the presence of goats had considerable effects upon the total percent cover and species composition of both the herbaceous and shrubby components of brushland areas (Coblentz, in press; Fig. 1). Here, the cover of all vegetation averaged over 50 % more (42 % compared with 27 %) in areas where goats had been eliminated about 15 years earlier. Additionally, California sagebrush (*Artemisia californica*), a favoured food of the goats, and major component of Catalina's shrubland areas, comprised more than one-fifth of the total percent cover in the goat-free areas, but was completely absent in the goat-inhabited area.

I also measured total herbaceous production on both goat-free and adjacent goat-inhabited areas on Catalina Island (Coblentz, in press). Total herbaceous production was nearly three times greater in the goat-free area during a drought year (2202 kg/ha compared with 818 kg/ha), and about 25 % greater during a wet year (2730 kg/ha compared with 2081 kg/ha). The higher productivity of vegetation in the

goat-free areas appeared related to greater amounts of vegetation remaining on the ground and acting as a moisture-retaining mulch, and the greater incorporation of organic matter into the soil. Percent utilisation of herbaceous vegetation in the goat-inhabited areas was higher in both years.

#### DISCUSSION

Goats are able to cause destruction of habitat because they are highly adaptable. Very little palatable forage escapes them; only plants growing in inaccessible

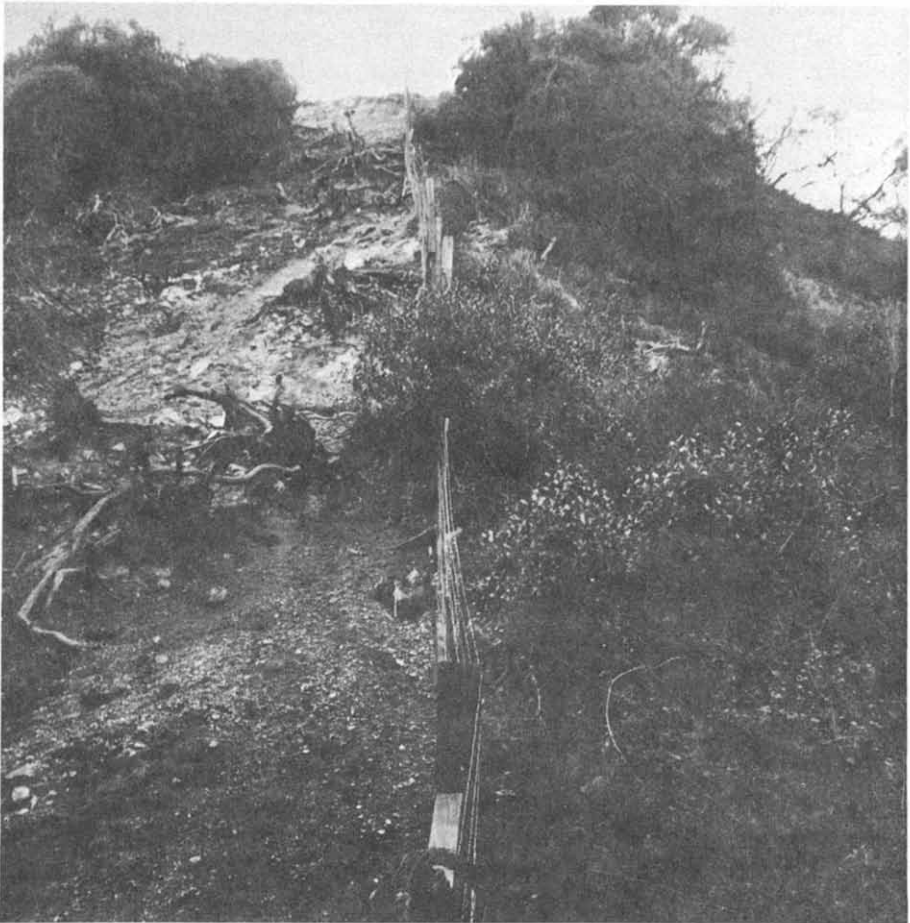


Fig. 1. View along fence separating goat-inhabited area from goat-free area near bottom of a brushy canyon. Goat-inhabited area is on the left, as indicated by many dead woody plants and a nearly complete lack of herbaceous cover. Santa Catalina Island, California.

locations are completely safe. Some plant species are avoided because they contain high levels of bitter oils or other compounds.

The feeding behaviour of the feral goat also results in serious disturbances to the vegetation. They sometimes strip most of the foliage off a particular shrub, thereby greatly weakening the individual plant. In addition, goats also damage taller shrubs and small trees by bending them down so that they might reach the upper foliage. Under normal circumstances, a plant able to grow above a certain height will have some degree of immunity from browsing; however, goats are able to climb into many low-growing trees and feed in the canopy.

Traditional behaviour within goat populations contributes to destruction of habitat. Goats tend to travel in single file when moving between two areas within their home range. Usually there are numerous established trails on any particular slope, and in fact the total land area taken up by trails may be as much as 1–2% (Coblentz, 1974). Vegetation is soon destroyed on trails, and so this land is essentially taken out of production (Fig. 2).

Goats tend to bed on certain exposed high points or cliffs within their home range, and individuals return to these sites regularly. The high amount of use such areas receive soon renders them barren, except for a few unpalatable plants. Serious erosion often begins in the vicinity of traditional bedding grounds.

Considering what little is known of the feral goat in insular ecosystems, it seems reasonable to ask, 'Why have feral animals been virtually ignored by ecologists?' The answer appears to be simply that the study of feral animals is not fashionable. Ecologists recognise that feral ungulates indeed do exist, and that where they do, they are a serious threat to both the native flora and fauna, and to the habitat in general. Wallace (1911) stated that goats were the 'greatest of all foes' to trees because of their ability to eat all available seedlings. Bates (1956), summed up both the state of our knowledge of feral animals, and the degree of attention that has been directed toward them by ecologists:

'The goat . . . might well be called the 'ecological dominant' over much of the Mediterranean region, the Venezuelan Andes, and many other parts of the world, including numerous oceanic islands. Yet, running through a series of ecology textbooks, I find no entry of 'Goat' in the indexes.

I was distressed to discover that in my own survey of a large sample of current ecology texts, goats have retained their anonymity.

In the past, problems caused by unchecked populations of feral ungulates have largely been acknowledged only in passing, as if such acknowledgement would suffice either to eliminate the problem, or to pass responsibility for solving it. As recently as 1966, in the published symposia of the Galapagos International Scientific Project, there were only five mentions of feral ungulates in the nearly 40 papers.

More recently, feral goats have been identified as severe threats to island ecosystems and in need of extensive and immediate study (Baker & Reeser, 1972;

Simkin *et al.*, 1972; Coblenz, 1976). Since feral animals are ecologically dominant in insular environments and have had such extensive effects, they should be a completely legitimate and fascinating subject of both basic and applied research.

In the Galapagos in 1835, Darwin first became convinced of the existence of



Fig. 2. Goat trails on Santa Catalina Island. One trail is considerably more worn than the others (main trail). As much as 2% of the total area may be out of production due to trails.

evolution (Huxley, 1966), largely because of the endemic species of finches (Geospizinae) and tortoises that he found on separate islands. It is interesting to consider that goats are evolving independently on those same islands at the present time, yet this phenomenon is occurring unnoticed!

## REFERENCES

- ATKINSON, I. A. E. (1964). Relations between feral goats and vegetation in New Zealand. *Proc. N. Z. ecol. Soc.*, **11**, 39–44.
- BAKER, J. K. & REESER, D. W. (1972). Goat management problems—Hawaii Volcanoes National Park. *US Department of the Interior, National Park Service Research Report*, No. 2, 63 pp.
- BATES, M. (1956). Man as an agent in the spread of organisms. In *Man's role in changing the face of the earth*, ed. by William L. Thomas Jr., 788–804. Chicago, University of Chicago Press.
- COBLENTZ, B. E. (1974). *Ecology, behavior, and range relationships of the feral goat*. Ph.D. Thesis, University of Michigan, Ann Arbor.
- COBLENTZ, B. E. (1976). *Report on the status and management of game animals, Santa Catalina Island, California*. Washington DC, Center for Natural Areas.
- COBLENTZ, B. E. (in press). Some range relationships of feral goats on Santa Catalina Island, California. *J. Range Mgmt.*
- DAWSON, E. Y. (1966). Cacti in the Galapagos Islands, with special reference to their relations with tortoises. In *The Galapagos: Proceedings of the symposia of the Galapagos International Science Project*, ed. by Robert I. Bowman, 209–14. Berkeley, University of California Press.
- DUFFEY, E. (1964). The terrestrial ecology of Ascension Island. *J. appl. Ecol.*, **1**, 219–51.
- DUNKLE, M. B. (1950). Plant ecology of the Channel Islands of California. *Allen Hancock Pacif. Exped.*, **13**, 247–86.
- GREENWAY, J. C. Jr. (1958). Extinct and vanishing birds of the world. *Spec. Publs Am. Comm. int. Wild Life Prot.*, No. 13, 518 pp.
- HAMANN, O. (1975). Vegetational changes in the Galapagos Islands during the period 1966–73. *Biol. Conserv.*, **7**, 37–59.
- HENDRICKSON, J. D. (1966). The Galapagos tortoises, *Geochelone Fitzinger 1835 (Testudo Linnaeus 1758 in Part)*. In *The Galapagos: Proceedings of the symposia of the Galapagos International Science Project*, ed. by Robert I. Bowman, 252–7. Berkeley, University of California Press.
- HUXLEY, J. (1966). Charles Darwin: Galapagos and after. In *The Galapagos: Proceedings of the symposia of the Galapagos International Science Project*, ed. by Robert I. Bowman, 3–9. Berkeley, University of California Press.
- JEWELL, P. A., MILNER, C. & BOYD, J. M. (eds.) (1974). *Island survivors: The ecology of the Soay sheep of St Kilda*. London, The Athlone Press.
- KOFORD, C. B. (1966). Economic resources of the Galapagos Islands. In *The Galapagos: Proceedings of the symposia of the Galapagos International Science Project*, ed. by Robert I. Bowman, 3–9. Berkeley, University of California Press.
- MACFARLAND, C. G. (1974a). The Galapagos giant tortoises (*Geochelone elephantopus*) Part I: Status of the surviving populations. *Biol. Conserv.*, **6**, 118–33.
- MACFARLAND, C. G. (1974b). The Galapagos giant tortoises (*Geochelone elephantopus*) Part II: Conservation methods. *Biol. Conserv.*, **6**, 198–212.
- MUELLER-DOMBOIS, D. & SPATZ, G. (1972). Study on the influence of introduced large herbivores on the vegetation in Hawaii Volcanoes National Park. *Int. Biol. Progr., Tech. Rep.*, No. 2, 42–9. Honolulu, University of Hawaii.
- RUDGE, M. R. (1976). Feral goats in New Zealand. In *The value of feral farm animals in New Zealand*, ed. by A. H. Whitaker and M. R. Rudge, 15–21. *N. Z. Dept. of Lands and Survey Inf. Series*, **1**, 1–84.
- SIMKIN, T., REEDER, W. G. & MACFARLAND, C. (1972). *Galapagos science: 1972 status and weeds. Report of the Galapagos Scientific Conference*. Washington DC, Smithsonian Institution.
- SPATZ, G. & MUELLER-DOMBOIS, D. (1973). The influence of feral goats on koa tree production in Hawaii Volcanoes National Park. *Ecology*, **54**, 870–6.
- SYKES, W. R. (1969). The effect of goats on vegetation on the Kermadec Islands. *Proc. N. Z. ecol. Soc.*, **16**, 13–16.
- THORNE, R. F. (1967). A flora of Santa Catalina Island, California. *Aliso*, **6**, 1–77.
- WALLACE, A. R. (1911). *Island life*, 3rd ed. London, Macmillan.
- WHITAKER, A. H. (1976). Feral sheep in New Zealand. In *The value of feral farm animals in New Zealand*, ed. by A. H. Whitaker and M. R. Rudge, 27–32. *N. Z. Dept. of Lands and Survey Inf. Series*, **1**, 1–84.
- WODZICKI, K. A. (1950). Introduced mammals of New Zealand. *N. Z. Dept. Sci. Ind. Res. Bull.*, No. 98, 255 pp.
- YOCOM, C. F. (1967). Ecology of feral goats in Haleakala National Park, Maui, Hawaii. *Am. Midl. Nat.*, **77**, 418–51.